

Project Plan for Mental Health Management System

Developed By: Afolabi Peter

Client:

Timeframe: 2 months

Total Cost: # 2,500,000

Key Features

1. Patient Features

- **Onboarding with Two-Factor Authentication (2FA):** Secure login using SMS and Email OTP for enhanced security.
- **Dashboard:** Visualize health statistics such as mood trends, symptom tracking, and progress over time.
- **Profile Management:** Update personal details, health history, and preferences.
- **Consultation Booking:** Schedule, reschedule, or cancel appointments with doctors.
- **Real-Time Communication:** In-app chat functionality for seamless communication with doctors, including document sharing.
- **Health Records History:** Access past consultation notes, prescriptions, and health data.
- **Health Tips:** Curated mental health tips and resources to support well-being.
- **Mood Tracking:** Log daily moods with customizable scales (e.g., 1-10 or emojis) and view trends via graphs.
- **Journaling:** A secure, private space for users to document thoughts, feelings, and reflections.
- **Self-Help Resources:** Access to articles, videos, audio content, and AI-powered insights for mental health support.

- **Crisis Support:** Quick access to emergency contact information (e.g., hotlines) and local mental health resources.
- **Notifications:** Timely reminders for appointments, medication, and system updates.

2. Doctor Features

- **Onboarding with 2FA:** Secure login using SMS and Email OTP for enhanced security.
- **Dashboard:** View consultation records, patient statistics, and performance metrics.
- **Profile Management:** Update professional details, availability, and consultation fees.
- **Consultation Integration:** Seamless integration with Zoom API for video consultations.
- **Real-Time Communication:** In-app chat functionality for communication with patients.
- **Health Records Access:** View patient history, progress, and treatment plans.
- **Notifications:** Reminders for upcoming appointments and system updates.
- **Treatment Planning:** Create, update, and share personalized treatment plans with patients.
- **Progress Monitoring:** Access patient mood logs, symptom tracking, and other health data to monitor progress.

3. Admin Features

- **Admin Dashboard:** Manage users (patients, doctors), view system analytics, and monitor usage.
- **Manual and Automatic Email System:** Send emails to users (e.g., reminders, updates).
- **Content Management:** Upload and manage health tips, articles, videos, and resources.
- **User Management:** Add, remove, or update user accounts.
- **Reporting and Analytics:** Generate insights on mental health trends and outcomes.

4. Technical Features

- **Cloud Hosting:** Web and Web APIs hosted on Digital Ocean.
- **Storage:** Amazon S3 for content storage (e.g., health records, images).

- **Mobile Apps:** Available on Playstore and Appstore.
- **SMS and Email Integration:** For OTP, notifications, and communication.
- **AI-Powered Insights:** Predictive analytics for identifying mental health risks.

Deliverables

1. Mobile Apps:

iOS (Appstore) and Android (Playstore) apps for patients and doctors.

2. Web Apps:

Responsive web applications for patients, doctors, and admins.

3. Admin Panel:

Full-featured admin dashboard for system management.

4. Code Sources:

Complete github source code for all platforms (mobile, web, admin).

5. Accounts:

Login credentials for all accounts created (hosting, Appstore, Playstore, Amazon S3, etc.).

Timeline

Mental Health Management System - Development Timeline

- **Team:** 1 Frontend Developer (React Native) & 1 Backend Developer (Node.js/PostgreSQL)
- **Timeline:** 2 Months (8 Weeks)

Month 1: Core Platform & Foundation

Week 1: Project Setup & Authentication

Frontend Team

- Initialize the React Native project for iOS, Android, and Web.
- Set up the core navigation structure for the app.
- Develop the Login and Signup screens.
- Build the 2FA OTP input flow (UI only).

Backend Team

- Set up the project on Digital Ocean.
- Design and create the core database schema (Users, Profiles).
- Build the complete Authentication API (Login, Signup, JWT tokens).
- Integrate Twilio (SMS) and SendGrid (Email) for sending OTPs.

Week 1 Goal: Frontend can make mock login calls, and the backend has a working /login endpoint.

Week 2: User Profiles & Dashboards

Frontend Team

- Build the Patient and Doctor Profile screens (both viewing and editing).
- Build the basic skeleton for the Patient and Doctor Dashboards.
- Implement the main tab navigation for both user types.

Backend Team

- Build the CRUD APIs for managing User Profiles.
- Develop the API endpoints that provide aggregated data for the dashboards.
- Set up file upload to Amazon S3 for profile pictures.

Week 2 Goal: The frontend can display real user profile data and placeholder dashboard stats from the backend.

Week 3: Core Patient Features

Frontend Team

- Build the Mood Tracking UI, including the form to log a mood and the history list.
- Build the Journaling UI, allowing users to create, edit, and view their entries.
- Build the screen to list Health Tips and Resources.

Backend Team

- Build APIs for creating, listing, and analyzing Mood Entries.
- Build the full set of APIs for Journal Entries (create, read, update, delete, list).
- Build the API for an Admin to create and manage Health Tips content.

Week 3 Goal: Patients can successfully log and view their moods and journal entries within the app.

Week 4: Admin & Core Systems

Frontend Team

- Build the Admin Dashboard with system statistics and user lists.
- Build the Admin screens for managing all users (view, add, deactivate).
- Build the Admin interface for uploading and managing Health Tips content.

Backend Team

- Build the comprehensive Admin APIs for user management.
- Finalize the APIs for the content management system.
- Begin building the architecture for the Notification system.

End of Month 1 Milestone: The core platform is functional. Patients, Doctors, and Admins can access their main screens and manage basic data.

Month 2: Advanced Features & Integration

Week 5: Booking & Real-Time Chat

Frontend Team

- Build the complete Appointment Booking flow (select doctor, choose time).
- Build the screens for viewing Upcoming and Past appointments.
- Build the user interface for the real-time chat, including the conversation list and message window.

Backend Team

- Build APIs for managing doctor availability and handling appointments (book, reschedule, cancel).
- Implement Socket.io to enable real-time, two-way chat.
- Build APIs to send, receive, and store the full chat history.

Week 5 Goal: Users can book appointments and see a functional, real-time chat interface.

Week 6: Video Consultation & Health Records

Frontend Team

- Integrate the Zoom SDK into the mobile and web apps.
- Build a "Join Consultation" screen that launches the Zoom meeting.
- Build the Health Records screen to display past consultation notes and prescriptions.

Backend Team

- Integrate the Zoom API to automatically generate meeting links for each appointment.
- Build APIs to store and retrieve consultation notes and prescriptions (Health Records).
- Connect the notification system to send reminders for upcoming appointments.

Week 6 Goal: A user can book an appointment and then seamlessly launch the Zoom video call directly from the app.

Week 7: Polish, AI & Analytics

Frontend Team

- Implement the Crisis Support screen with emergency resources.
- Build charts and graphs to visualize Mood Trends.
- Integrate and display the AI-Powered Insights on the dashboard.
- Begin cross-platform testing and bug fixing on iOS, Android, and Web.

Backend Team

- Develop and integrate the AI-Powered Insights engine (e.g., for analyzing mood trends).
- Build the final APIs for admin analytics and reporting.
- Finalize all automated notification triggers.

Week 7 Goal: Feature Freeze. All planned features are implemented, integrated, and ready for final testing.

Week 8: Testing, Deployment & Launch

Frontend Team

- Intensive bug fixing, performance optimization, and UI polish.
- Final testing on all target devices and platforms.
- Build production-ready app bundles (APK/IPA) and web bundles.
- Submit the final applications to the Apple App Store and Google Play Store.

Backend Team

- Deploy the final backend API to the production Digital Ocean server.
- Perform final database migrations and security checks.
- Assist the frontend team with any last-minute API issues.
- Prepare final technical documentation for handover.

End of Month 2 Milestone: System is live. Apps are submitted to stores, web app is hosted, and final code and documentation are delivered to the client.

Payment Schedule

Total Project Cost: ₦2,500,000

The payment schedule is designed to align with key project milestones, ensuring shared commitment and progress.

Milestone 1: Project Kick-off & Mobilization

Payment: ₦750,000 (30%)

Trigger: Upon project initiation and signing of the agreement. This initial payment secures the development team and resources for the project.

Milestone 2: Core Platform Completion & Mid-Project Review

Payment: ₦750,000 (30%)

Trigger: Upon successful completion and demonstration of the fully integrated core platform at the end of Month 1. This includes all features from the first month's timeline (Authentication, User Profiles, Dashboards, Mood Tracking, Journaling, and Admin Management).

Milestone 3: Final Delivery, Testing & Project Launch

Payment: ₦1,000,000 (40%)

Trigger: Upon final delivery and launch, which includes:

- Completion of all advanced features (Booking, Chat, Video, AI Insights).
- Successful testing and bug fixing.
- Submission of mobile apps to the Play Store and App Store.
- Full handover of all source code, documentation, and account credentials.

Ongoing Costs (Post-Delivery)

The ongoing costs for maintaining the Mental Health Management System are broken down into essential and optional services. The final cost will directly depend on your number of active users and platform usage.

Essential Infrastructure (Required for the app to function):

1. **Cloud Hosting & Servers (Digital Ocean):** This is the core cost for running your web apps, admin panel, and backend API. The estimated monthly cost is between **₦40,000 and ₦80,000**. This is a variable cost that will increase as your user base grows.
2. **File Storage (Amazon S3):** This covers the secure storage of all user-uploaded files, such as profile pictures and documents shared in chats. The estimated monthly cost is between **₦10,000 and ₦30,000**.
3. **Communication Services (Twilio for SMS & SendGrid for Email):** This is for sending all automated OTPs, appointment reminders, and system notifications. The cost is highly usage-based, with an estimated range of **₦15,000 to ₦50,000 per month**.
4. **Database Management & Backups:** This ensures your patient and application data is stored reliably and is regularly backed up to prevent loss. The estimated cost for a managed database service is approximately **₦25,000 per month**.

Total Estimated Monthly Cost for Essential Infrastructure: **₦90,000 to ₦185,000**.

Additional Optional Service

AI/ML Service (For AI-Powered Insights): If you actively use the AI feature for analyzing patient mood trends and generating insights, this will be an additional cost. The estimated range is **₦10,000 to ₦50,000 per month**, depending on how frequently the feature is used.

Grand Total Estimated Monthly Cost: **₦100,000 to ₦235,000**.

Post-Delivery Support

This outlines the support and maintenance services provided after the final system handover and launch.

Support Duration: 3 Months (included in the project cost)

What is Included (Scope of Support)

1. Bug Fixing and Error Resolution:

- We will address and fix any software bugs, errors, or unexpected crashes that occur in the delivered system.
- This ensures the application runs smoothly and as intended.

2. Critical Security Updates

We will apply necessary patches to the core framework (React Native, Node.js) and dependencies to address any critical security vulnerabilities that are discovered.

3. Third-Party Service Compatibility

We will ensure the application remains compatible with essential integrated services (e.g., Zoom API, Twilio, SendGrid) if those services update their APIs in a way that breaks current functionality.

4. Minor Adjustments

We will handle very minor tweaks to the user interface (UI) or user experience (UX) based on initial user feedback, subject to a mutual agreement on the scope. (e.g., changing label text, adjusting color shades).

Support Service Level Agreement (SLA)

- **Critical Bugs** (System down, major feature broken): Initial response within 24-48 hours.
- **Minor Bugs** (Non-critical UI issues, small errors): Resolution within 3-5 business days.

What is Not Included

The following services are NOT covered under the standard post-delivery support and would require a separate support retainer or a new project agreement:

1. **Development of New Features:** Any requests for new functionality, modules, or significant changes to the existing system.
2. **Major Platform Upgrades:** Upgrading to major new versions of React Native or other core technologies that require significant development work.
3. **Data Migration or Integration:** Assistance with importing existing data from other systems or integrating with new third-party services not in the original plan.
4. **Content Creation & Upload:** Writing new health tips, articles, or managing the content within the admin panel.
5. **Server Management & Performance Optimization:** Proactive server monitoring, performance tuning, and handling issues related to server configuration beyond the initial setup.

Extended Support Plans (After 3 Months)

After the initial 3-month support period ends, we offer ongoing retainer plans to ensure the long-term health and evolution of your application. These can be customized but generally include:

- **Basic Plan:** Continued bug fixes and security updates.

- **Standard Plan:** Basic plan + a monthly allocation of hours for small feature additions or changes.
- **Premium Plan:** Standard plan + proactive monitoring, performance optimization, and dedicated support.

A detailed proposal for extended support will be provided before the initial 3-month period concludes.